Name

Mathematics Extension 1 Year 12

Central Limit Theorem

Book 1

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Syllabus Content

ME1-12-06 solves problems involving binomial distributions, sampling distribution of the mean and the central limit theorem

Sampling distribution of the mean and the central limit theorem

- Define a statistical population as the entire group of people or objects about which information is sought
- Define a sample as a selection of people or objects drawn from a population
- Recognise that a statistic taken from a sample will be a good approximation if the sample is random and the sample size is large enough, typically $n \geq 30$
- Recognise that, in general, the distribution of a population and a statistic, such as its mean, are unknown
- Recognise that it is likely that the distribution of a random sample will resemble the distribution of the population when the sample size, n, is large enough, typically $n \geq 30$
- Define the sample mean for a sample $X_1, X_2, \ldots, X_n$ taken from a population as $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- Recognise that the sample means obtained from repeated sampling may be different, even if all the samples are of the same size
- Define, for a given sample size n, the sampling distribution of the mean to be the distribution of the sample means of all samples of size n
- Recognise that sample mean, $\bar{X}$, is a random variable that estimates the population mean, μ, when the size of the sample is large enough and consider its usefulness in making predictions in a variety of contexts including politics, finance, agriculture and biology
- State the central limit theorem as: for a population with mean μ and variance σ^2, provided the sample size n is large enough ($n \geq 30$), the sampling distribution is approximately normal with mean μ and variance $\frac{\sigma^2}{n}$; that is, $\bar{X}$ is approximately $N\left(\mu, \frac{\sigma^2}{n}\right)$
- Recognise the significance of the central limit theorem for populations that are not necessarily normally distributed; that, irrespective of the population distribution, for a sufficiently large sample size, the sampling distribution of the mean is approximately normal
- Use the formulas $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ as the mean and the variance of the sampling distribution of the mean for samples of size n drawn from a population with mean μ and variance σ^2
- Examine the effect of the sample size on the variance of sample means with digital tools
- Apply the central limit theorem to estimate the probability that the sample mean lies within given bounds

The Sample Proportion

Investigation The sample proportion

Imagine that a city has a population of 1 million people, with 40% owning a bicycle.

You repeatedly survey random samples of 1000 people and record the results.

Let X be the number of bicycle owners in a sample. Since each person either owns a bicycle or doesn't, X is a …… random variable.

Calculate the proportion $\hat{p}$ of people in each sample who own a bike.

Sample A: 437 out of 1000 people own a bike. $\hat{p}_A =$ ……

Sample B: 412 out of 1000 people own a bike. $\hat{p}_B =$ ……

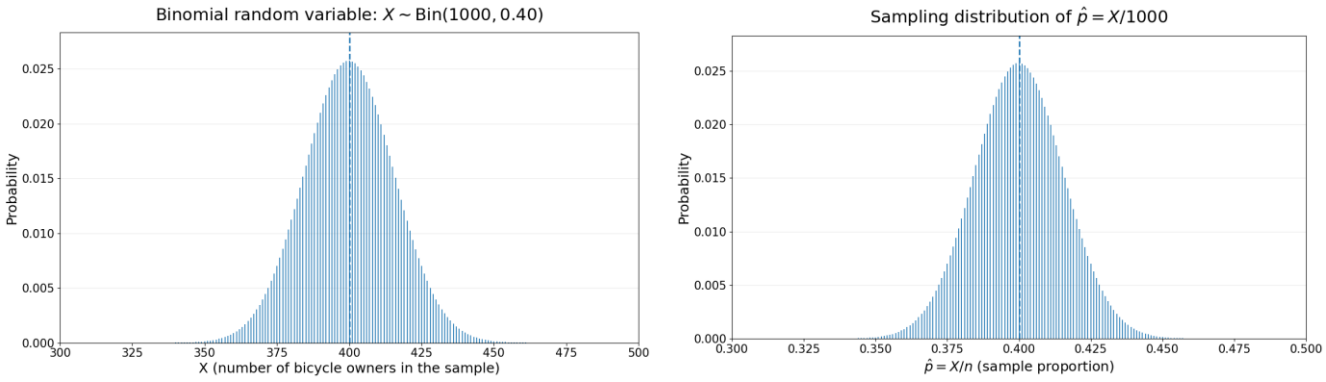
Sample C: 385 out of 1000 people own a bike. $\hat{p}_C =$ ……

The **sample proportion** $\hat{p}$ is the number of successes recorded in a sample ($X \sim \text{Bin}(n, p)$), divided by the sample size n.

$$\hat{p} = \frac{X}{n}$$

The distribution of $\hat{p}$ is just the distribution of X horizontally dilated by factor $\frac{1}{n}$.

Notice that the probabilities are the same $P\left(\hat{p} = \frac{r}{n}\right) = P(X = r) = \binom{n}{r} p^r q^{n-r}$ since we are relabelling each x value: 400 ↦ 0.4.



If we take lots of samples and average the results, we get a good estimate of the true probability (**law of large numbers**) and for sufficiently large numbers of trials, the distribution of sample proportions becomes approximately normal (**central limit theorem**).

Statisticians usually work with sample proportions because proportions can be compared across samples of different sizes and directly estimate a population percentage. Saying "40% of people own a bicycle" is generally more useful than saying "400 people own a bicycle", because the second statement depends on the sample size.

Sample Proportions

The **sample proportion** $\hat{p}$ is the number of successes recorded in a sample ($X \sim \text{Bin}(n, p)$), divided by the sample size n.

$$\hat{p} = \frac{X}{n}$$

Expected Value	Variance	Standard Deviation
$E(\hat{p}) = E\left(\frac{X}{n}\right)$	$\text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right)$	$\sigma(\hat{p}) = \sqrt{\text{Var}(X)}$
$= \frac{E(X)}{n}$	$= \frac{E(X)}{n^2}$	$= \sqrt{\frac{np(1-p)}{n^2}}$
$= \frac{np}{n}$	$= \frac{np(1-p)}{n^2}$	$= \frac{\sqrt{np(1-p)}}{n}$
$= p$	$= \frac{p(1-p)}{n}$	

Example Calculate $E(\hat{p}), \text{Var}(\hat{p})$ and $\sigma(\hat{p})$

A factory produces circuit boards. In the long run, 30% of boards are faulty. A random sample of 150 boards is taken. Let X be the number of faulty boards in the sample, and let $\hat{p} = \frac{X}{150}$

Find $E(\hat{p})$ and $\sigma(\hat{p})$

$X \sim \text{Bin}(150, 0.3)$

$E(\hat{p}) = p = 0.3$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} = \frac{0.3(1-0.3)}{150} = 0.0014$$

$$\sigma(\hat{p}) = \sqrt{Var(\hat{p})} = \sqrt{0.0014}$$

Guided Practice

12% of all emails arriving at a large server are spam. A random sample of 300 emails is taken. Let $\hat{p}$ be the proportion of spam emails in the sample. Find $E(\hat{p})$ and $\text{sd}(\hat{p})$, correct to four decimal places

$E(\hat{p}) = 0.12$

$\text{sd}(\hat{p}) = \sqrt{0.000352} = 0.0188$

In the factory circuit board example, $X \sim \text{Bin}(n, 0.3)$ and $\hat{p} = \frac{X}{n}$, what is the smallest sample size n for which $\sigma(\hat{p}) < 0.01$?

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a A student tosses five coins and the results are H T H T T.
What is the sample proportion $\hat{p}$ of heads in this experiment?

$\hat{p} = \frac{2}{5}$

b A student selects ten cards from a standard pack with replacement, and records the suits:

$$\heartsuit, \clubsuit, \spadesuit, \diamondsuit, \diamondsuit, \spadesuit, \heartsuit, \spadesuit, \spadesuit, \diamondsuit$$

What is the sample proportion $\hat{p}$ of spades in this sample?

$\hat{p} = \frac{4}{10}$

c A manufacturer takes a sample of 12 items from a recent batch of gizmos, testing each item to see if it passes quality control, P, or not, F.
The results were P P P F F P P P P F P P.
What is the proportion $\hat{p}$ of items that pass?

$\hat{p} = \frac{9}{12} = \frac{3}{4}$

2 Find the sample proportion $\hat{p}$ for each situation described.

a Out of a sample of 2150 people, 178 were found to be left-handed.

$\hat{p} = \frac{178}{2150} = \frac{89}{1075}$

b Just 7 cars had defective handbrakes out of a sample of 800.

$\hat{p} = \frac{7}{800}$

c In a sample of 520 SIM cards, 25 were found to be faulty.

$\hat{p} = \frac{25}{520} = \frac{5}{104}$

d There were 58 tagged pigeons in a sample of 295.

$\hat{p} = \frac{58}{295}$

3 Find the mean and variance of each distribution of the sample proportion, given:

a $p = 0.63, n = 30$

$\mu = 0.63$
$\sigma^2 = \frac{0.63(0.37)}{30} = 0.00777$

b $p = 0.21, n = 55$

$\mu = 0.21$
$\sigma^2 = \frac{0.21(0.79)}{55} = 0.0030$

c $p = 0.7, n = 48$

$\mu = 0.7$
$\sigma^2 = \frac{0.7(0.3)}{48} = 0.0044$

d $p = 0.42, n = 76$

$\mu = 0.42$
$\sigma^2 = \frac{0.42(0.58)}{76} = 0.0032$

e $p = 0.55, n = 105$

$\mu = 0.55$
$\sigma^2 = \frac{0.55(0.45)}{105} = 0.0024$

4 A single fair coin is tossed five times, and the number x of heads is recorded.

x	0	1	2	3	4	5
$P(X = x)$						

Let $X \sim B(5, \frac{1}{2})$.

x	0	1	2	3	4	5
$P(X = x)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$
$\hat{p}$	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1
$P(\hat{p})$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

a Complete the table using the binomial probability formula.

b Convert part a to a table of probabilities of the sample proportions.
Divide each score x by 5 to obtain the corresponding sample proportion.

c Calculate the mean of the second table.

d How would you interpret this mean?

$E(\hat{p}) = \sum \hat{p} P(\hat{p}) = 0.5$

The mean equals the probability $p = 0.5$ of heads in each Bernoulli trial.

5 Every Saturday for 20 weeks, a marketer surveyed five people chosen at random in a suburban shopping centre. The first question is, 'Do you live in the suburb?', and the weekly frequency of the number x of Yes answers is given in the table.

x	0	1	2	3	4	5
Frequency f	1	1	3	2	6	7

a Complete a table in which the rows are the sample proportions $\hat{p}$ and the relative frequencies f_r.

b Calculate the mean of this second table.

c What is this mean an estimate of?

$\hat{p}$	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1
f_r	$\frac{1}{20}$	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{2}{20}$	$\frac{6}{20}$	$\frac{7}{20}$

$\bar{\hat{p}} = \sum \hat{p} f_r = 0.72$

It estimates the probability that a shopper selected at random lives in the suburb.